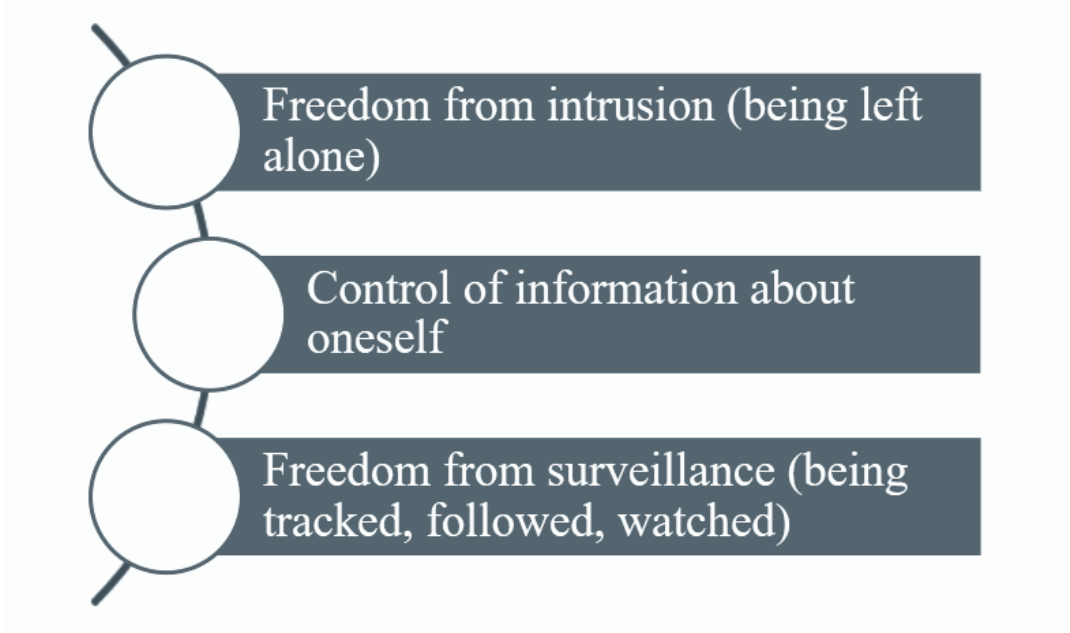
A GIFT OF FIRE THIRD EDITION - SARA BAASE
CHAPTER 2: PRIVACY

Lecture 1
Privacy and Computer Technology

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PRIVACY AND COMPUTER TECHNOLOGY

- Three Key Aspects of Privacy:



PRIVACY RISKS AND PRINCIPLES

Privacy threats come in several categories:

- Intentional, institutional uses of personal information
- Unauthorized use or release by “insiders”
- Theft of information
- Inadvertent leakage of information
- Our own actions

PRIVACY AND COMPUTER TECHNOLOGY

New Technology, New Risks:

- Government and private databases
  - Thousands of databases containing personal information about us.
  - Profiles of our personal information could be created easily.
- Sophisticated tools for surveillance and data analysis
  - Cameras, GPS, cell phones
- Vulnerability of data
  - Leaks of data happens, existence of data presents a risk.

PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Examples:

- Search query data
  - Search engines collect many terabytes of data daily.
  - Data is analyzed to target advertising and develop new services.
  - Who gets to see this data? Why should we care?
- Smartphones
- Location apps
- Data sometimes stored and sent without user's knowledge

PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Summary of Risks:

- Anything we do in cyberspace is recorded.
- Huge amounts of data are stored.
- People are not aware of collection of data. Software is complex.
- Leaks happen.
- A collection of small items can provide a detailed picture.
- Re-identification has become much easier due to the quantity of information and power of data search and analysis tools.

PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Summary of Risks:

- If information is on a public Web site, it is available to everyone.
- Information on the Internet seems to last forever.
- Data collected for one purpose will find other uses.
- Government can request sensitive personal data held by businesses or organizations.
- We cannot directly protect information about ourselves. We depend upon businesses and organizations to protect it.

PRIVACY AND COMPUTER TECHNOLOGY (CONT.)

Terminology and principles for data collection and use:

- Invisible information gathering
  - Collection of personal information about someone without the person's knowledge
  - Unauthorized software, Cookies, ISP providers •
- Secondary use
  - Use of personal information for a purpose other than the one it was provided for .
  - Data mining
    - Searching and analyzing masses of data to find patterns and develop new information or knowledge
  - Computer matching
    - Combining and comparing information from different databases (using social security number, for example, to match records)

Computer profiling

- Using data in computer files to predict likely behaviors of people
- Analyzing data in computer files to determine characteristics of people most likely to engage in certain behavior
- Businesses find new consumers
- Government detects fraud and crime activities
- Businesses engage in profiling to determine consumer propensity toward a product or service.
- Government agencies use profiling to create descriptions of possible terrorist

Principles for Data Collection and Use:

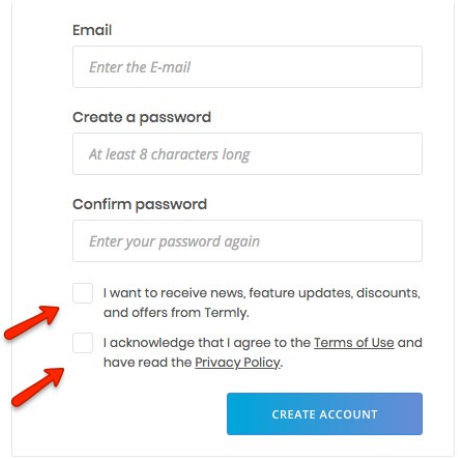
- Informed consent
- Opt-in and opt-out policies
- Fair Information Principles (or Practices)
- Data retention - Data retention is the storing of information for a specified period.

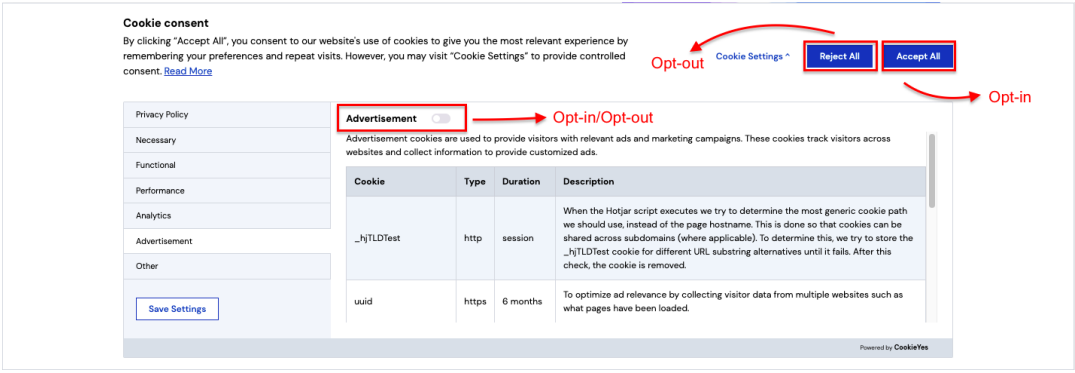
Principles for Data Collection and Use:

- Informed consent :
  - People should be informed about the data collection and use policies of a business or organization,
  - They can then decide whether or not to interact with that business or organization.
- Opt-in and opt-out policies :
  - Opt-in :
    - Opting in means that a user will take an affirmative action to offer their consent.
    - The most common way businesses implement opt-in methods is through checkboxes. When presented with a checkbox, the user must take action to check the box, which denotes their consent.

Principles for Data Collection and Use:

- Opt-in policies:
  - Opting in can be used in a variety of situations, including subscribing to email and newsletter mailing lists, accepting cookie use, and agreeing to legal policies.





Principles for Data Collection and Use:

- Opt-out policies :
  - Opting out means a user takes action to withdraw their consent.
  - Not choosing to subscribe to newsletters, unticking a previously ticked checkbox, not consenting to save personal details, rejecting the use of cookies, etc. are some examples of opt-out.



Principles for Data Collection and Use:

- Fair Information Principles (or Practices)
  - Inform people when you collect information about them, what you collect, and how you use it.
  - Collect only data needed
  - Offer a way for people to opt out from mailing lists, advertising, and other secondary uses.
  - Offer a way for people to opt out from features and services that expose personal information.
  - Keep data only as long as needed
  - Protect security of data (from theft and from accidental leaks). Provide stronger protection for sensitive data.
  - Policies for responding to law enforcement

Digital Privacy: Navigating Risks and Ethical Principles

THE LANDSCAPE OF MODERN PRIVACY RISKS



Freedom from Intrusion
The right to be free from unauthorized interference.

Control of Information
The right to maintain control over personal data.

Freedom from Surveillance
The right to be free from unauthorized monitoring.

THE THREE PILLARS OF PRIVACY



THE PERMANENT CYBERSPACE RECORD

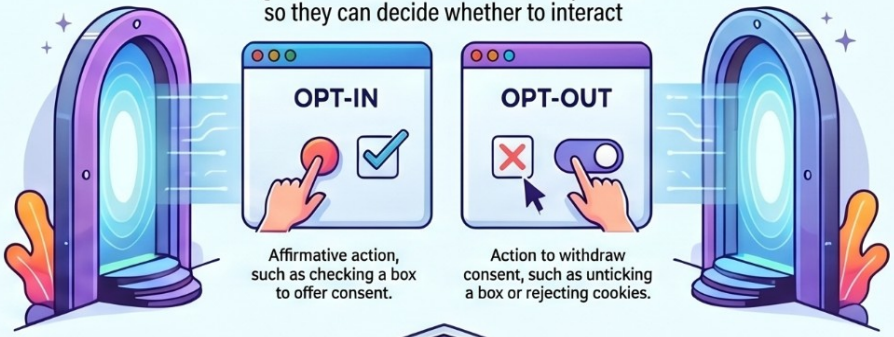
Digital activities are recorded and stored indefinitely, making re-identification easier through data analysis.

SECONDARY USE AND DATA MINING

Personal information is frequently analyzed to find patterns for purposes beyond the original intent.

PRINCIPLES FOR ETHICAL DATA PROTECTION

Organizations must inform users of policies so they can decide whether to interact



Affirmative action, such as checking a box to offer consent.

Action to withdraw consent, such as unticking a box or rejecting cookies.



COLLECT ONLY NECESSARY DATA

PROTECT FROM LEAKS

DELETE WHEN NO LONGER NEEDED

FAIR INFORMATION PRACTICES (FIP)

Collect only necessary data, protect it from leaks, and delete it when no longer needed.

NotebookLM

Thank You